

Minghui Liu

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Research Focus: Post-training and evaluation for reasoning and agentic language models

EDUCATION

University of Maryland, College Park

Ph.D. Candidate, Computer Science.

Advisor: Prof. Furong Huang

Jan. 2021 – Present

Trinity College, Hartford, CT

B.S. in Computer Science, summa cum laude with distinction; Phi Beta Kappa.

Sep. 2013 – May 2017

Top 3%

SELECTED RESEARCH

* Equal contribution

β -OPSD: Deriving with Policy Optimization, Training with Self-Distillation

J. Xu, M. Liu*, J. Zhang, T. Goldstein, F. Huang*

[arXiv 2026](#)

- Derived β -OPSD, a policy-optimization family whose closed-form optimum geometrically interpolates between a reference policy and a privileged teacher.
- Converted the optimum into an efficient mixed-logit distillation target with return-to-go credit assignment, improving mathematical reasoning and training stability over vanilla OPSD.

Scheduling Thoughts: Learning the Order of Thought in Diffusion Language Models

J. Xu, M. Liu, A. Agrawal, Y. Chen, F. Huang

[ICML 2026](#)

- Formulated token-unmasking order as policy optimization, deriving a tractable KL-based dense reward and training lightweight GRPO schedulers with frozen diffusion language models.
- Raised Sudoku accuracy from 82.0% to 91.8% and GSM8K pass@1 from 64% to 76%, consistently matching or exceeding heuristic schedules.

VeriGate: Verifier-Gated Step-Level Supervision for GRPO

A. Agrawal, M. Liu*, F. Huang*

[arXiv 2026](#)

- Developed verifier-gated GRPO with continuation-aware token-level credit assignment, activating process supervision only when outcome rewards provide no learning signal.
- Reduced zero-gradient failures and reward hacking, improving average accuracy by 20% and 12% for 1.5B and 7B models across six reasoning benchmarks.

Agentic Critical Training

W. Liu, M. Liu, S.-T. Ho, S. Chakraborty, X. Wang, F. Huang

[arXiv 2026](#)

- Designed an RL objective that trains agents to compare alternative actions and develop autonomous self-reflection rather than imitate generated critiques.
- Improved performance across three agent benchmarks by 5.07 points over imitation learning and 2.42 points over reflection distillation, with strong out-of-distribution transfer.

Ladders of Thought: A Self-Evolving Curriculum of Progressively Simplified Reasoning Traces

M. Liu, T. Magelinski, D. Yuan, Q. Yu, F. Huang

[arXiv 2025](#)

- Created faithful, progressively simplified reasoning tasks and a self-evolving bandit curriculum that adapts training difficulty to model learning progress.
- Improved small-LLM accuracy by up to 32 percentage points on arithmetic and 25 points on multi-hop reasoning while accelerating convergence and out-of-distribution generalization.

When AI Reviews Train AI Reviewers: Scientific-Judgment Collapse and Mitigation

S.-T. Ho, M. Liu*, F. Huang*

[arXiv 2026](#) | [Code](#) | [Project](#)

- Identified scientific-judgment collapse in recursive AI-reviewer training, including 11% lower same-paper and 5% lower corpus-level semantic diversity under fully synthetic exposure.
- Developed TrustReviewer using a curated 112,743-example, 1.9B-token corpus and paired activation steering, achieving 75.4% recommendation exact match versus 73.1% for OpenReviewer while improving rating diversity.

HashEvict: A Pre-Attention KV Cache Eviction Strategy using Locality-Sensitive Hashing
M. Liu, T. Rabbani, T. O'Halloran, A. Sankaralingam, C. Fermüller, Y. Aloimonos, M. A. Hartley, F. Huang, [arXiv 2024](#)

- Designed dynamic pre-attention KV-cache eviction using SimHash projections and a lightweight binary structure, avoiding attention-score computation during token selection.
- Reduced KV-cache size by 30–70% while preserving performance across reasoning, retrieval, and summarization tasks.

WORK EXPERIENCE

Applied Research Intern
Capital One, New York, NY

June 2026 – Aug 2026

- Investigated on-policy distillation in representation space, examining how different forms of privileged information affect student learning.
- Designed controlled experiments comparing privileged signals and representation-level objectives, evaluating their effects on optimization dynamics and downstream performance.

Applied Research Intern
Capital One, New York, NY

June 2025 – Aug 2025

- Developed a self-evolving curriculum that generates progressively simplified reasoning tasks and adapts difficulty online for small LLMs.
- Designed experiments demonstrating up to 32-point accuracy gains, faster convergence, and stronger out-of-distribution generalization.

HPC Software Engineer
Altair Engineering, Sunnyvale, CA

Aug 2017 – Jan 2021
[PROJECT GITHUB](#)

- Enhanced OpenPBS job scheduler accuracy, added Docker support, and optimized Linux build process.

ACADEMIC SERVICE

Gold Reviewer, ICML 2026 (top 20%) | Reviewer, NeurIPS 2026; AAAI 2027; ICLR 2027

SKILLS

<i>LLM Post-Training</i>	PyTorch, Hugging Face Transformers, TRL, PEFT/LoRA, LlamaFactory, verl, GRPO/RLVR, knowledge distillation
<i>Training & Inference Systems</i>	vLLM, DeepSpeed, FSDP, Accelerate, FlashAttention-2, Weights & Biases, Slurm, distributed training
<i>Programming & HPC</i>	Python, C, CUDA, Cython, MPI, OpenMP, Linux, Docker